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Automation of the Waterspout Nomogram

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Introduction

Purpose

- To develop an algorithm that generates a waterspout prognostic field for the Great Lakes

Advantages

- Dramatically reduces diagnosis time of waterspout potential
- More efficient coordination between forecast offices
- Precursor upstream forecast events viewable (a better spatial sense)

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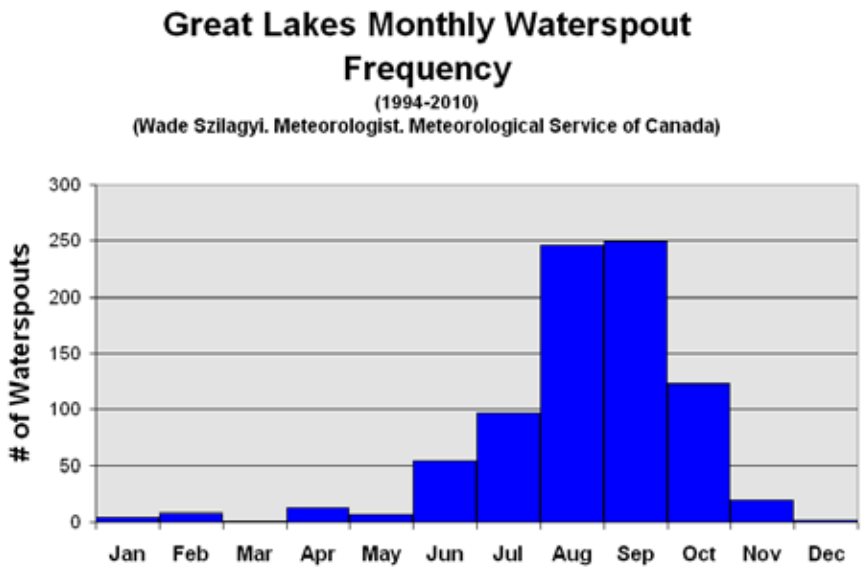
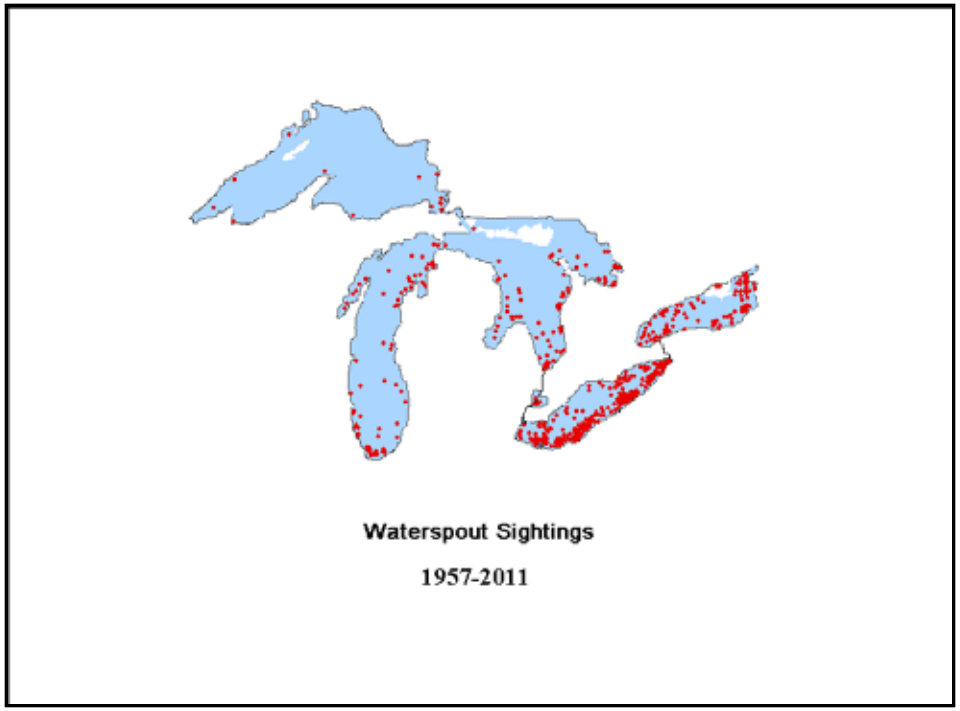
Waterspout Climatology over the Great Lakes





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Development History of the Waterspout Nomogram

- 1994 – Intensive investigation initiated into waterspout activity over the Great Lakes
- 1996 – Waterspout Nomogram
- 2004 – Szilagyi Waterspout Index (-10 to +10)
- 2011 – Experimental Waterspout Prognostic System

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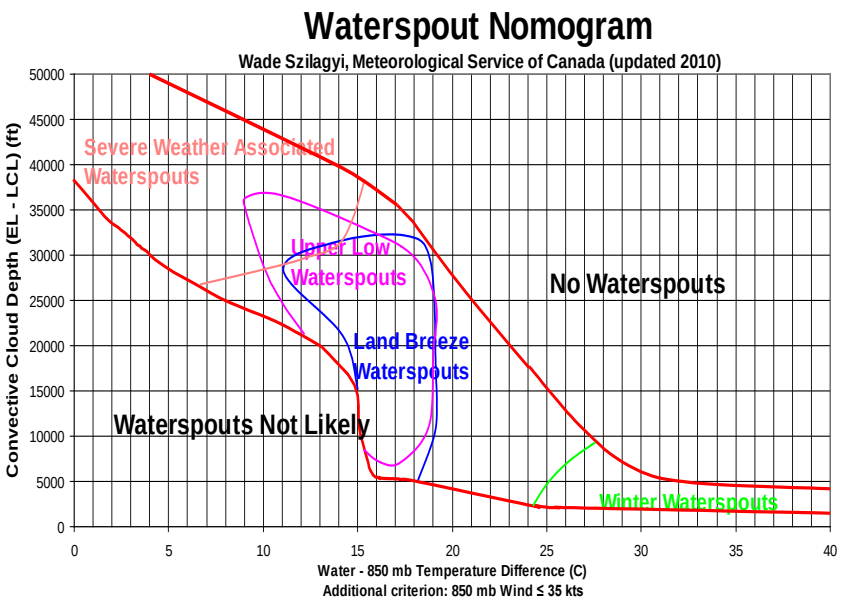
Waterspout Nomogram

- An empirical technique to forecast waterspouts
- Based on 207 events over the Great Lakes from 1988 to 2011
- Predictors:
  1. Water-850 mb temperature difference (ΔT)
  2. Convective cloud depth (EL - LCL = ΔZ)
  3. 850 mb wind speed (U_{850})



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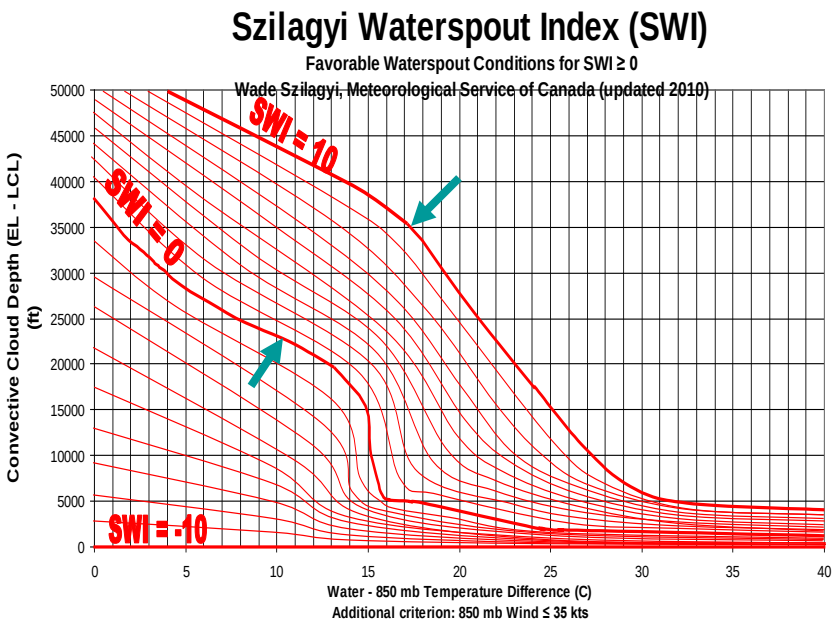
Szilagyi Waterspout Index (SWI)

- Quantifies the likelihood of waterspout formation
- Based on the Waterspout Nomogram
- A set of dimensionless SWI values (from -10 to +10) is plotted on the Waterspout Nomogram
- Waterspouts are likely to occur when SWI ≥ 0. The larger the SWI the higher the potential
- SWI is a function of both ΔT and ΔZ



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A New Waterspout Prognostic System

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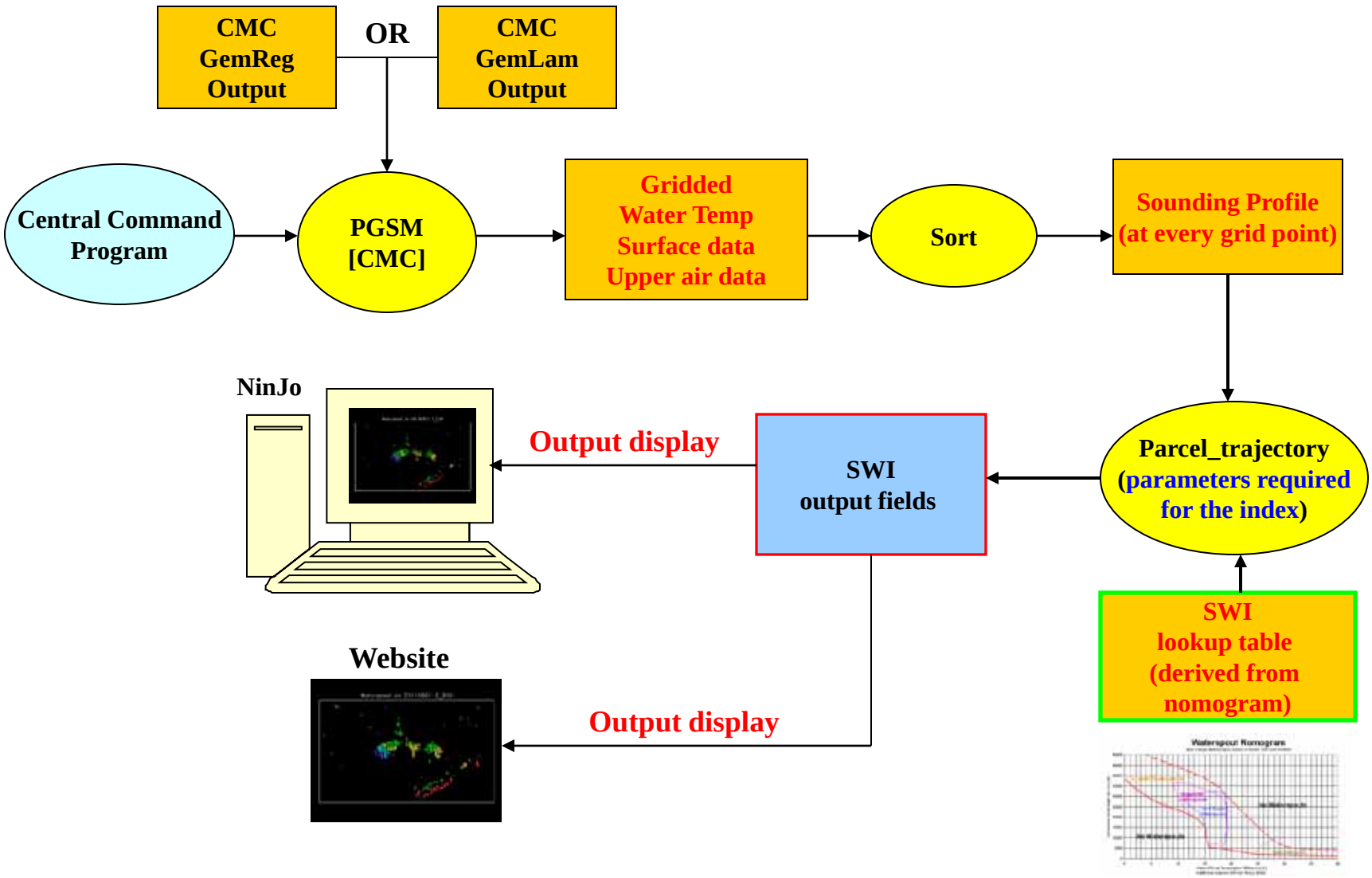
An Overview of the Waterspout Prognostic System



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Example

Case 1

August 21, 2011 – The Goderich Tornado Event

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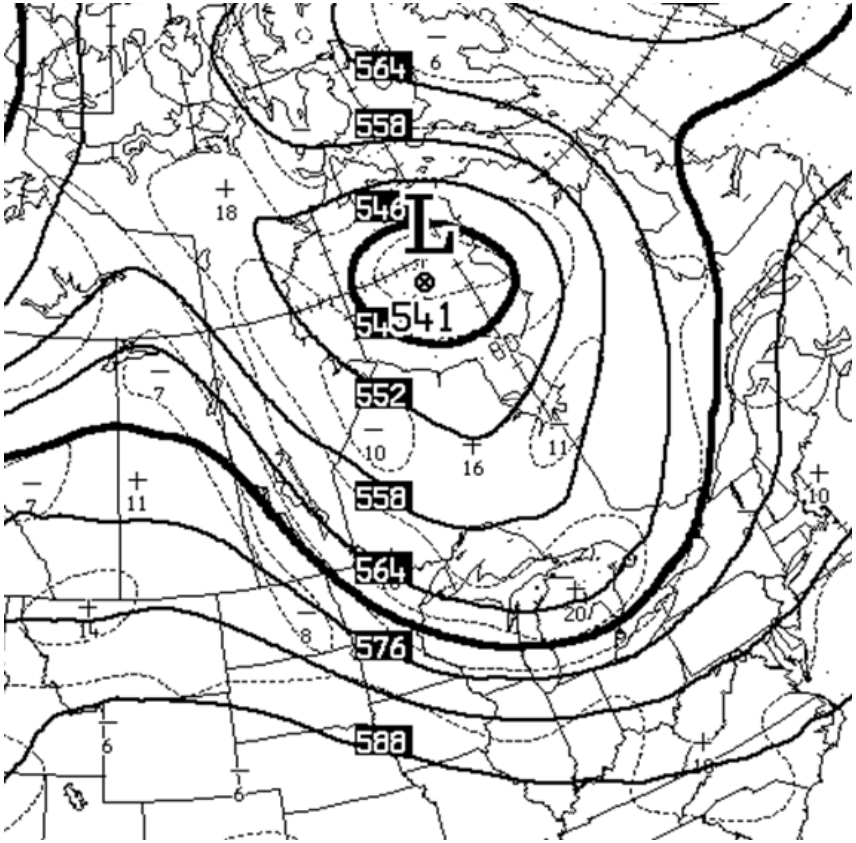
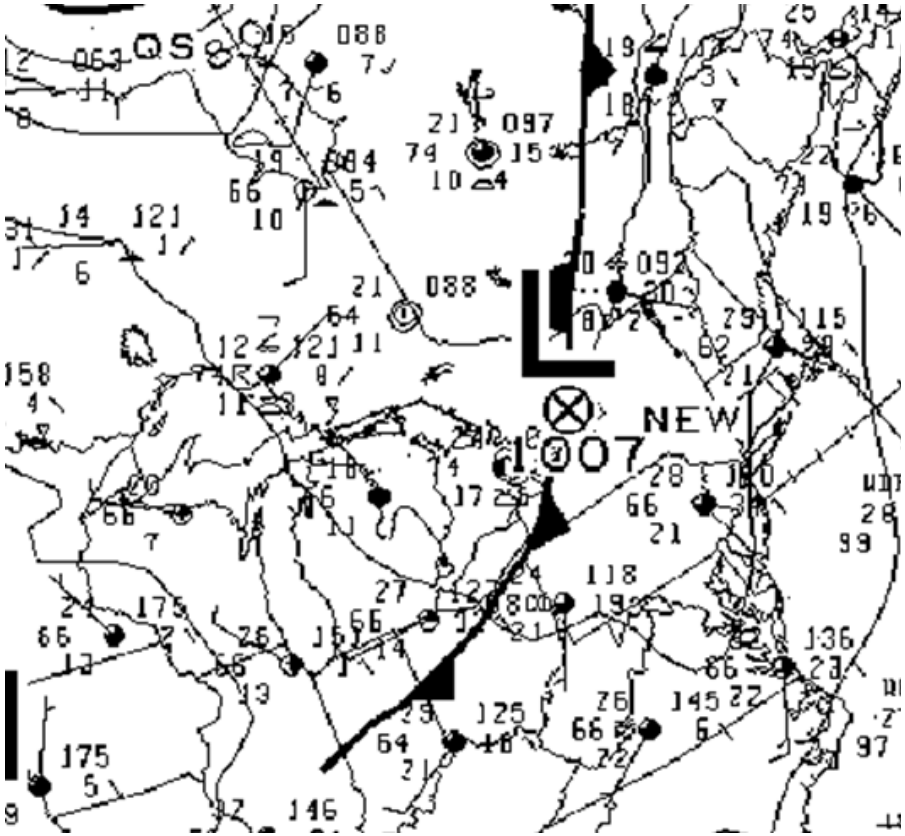
GEMREG model output for 500 mb height and surface analysis 18Z Aug 21, 2011

500 mb Height / Vorticity
Valid 18Z 2011/08/21



Surface analysis
Valid 18Z 2011/08/21



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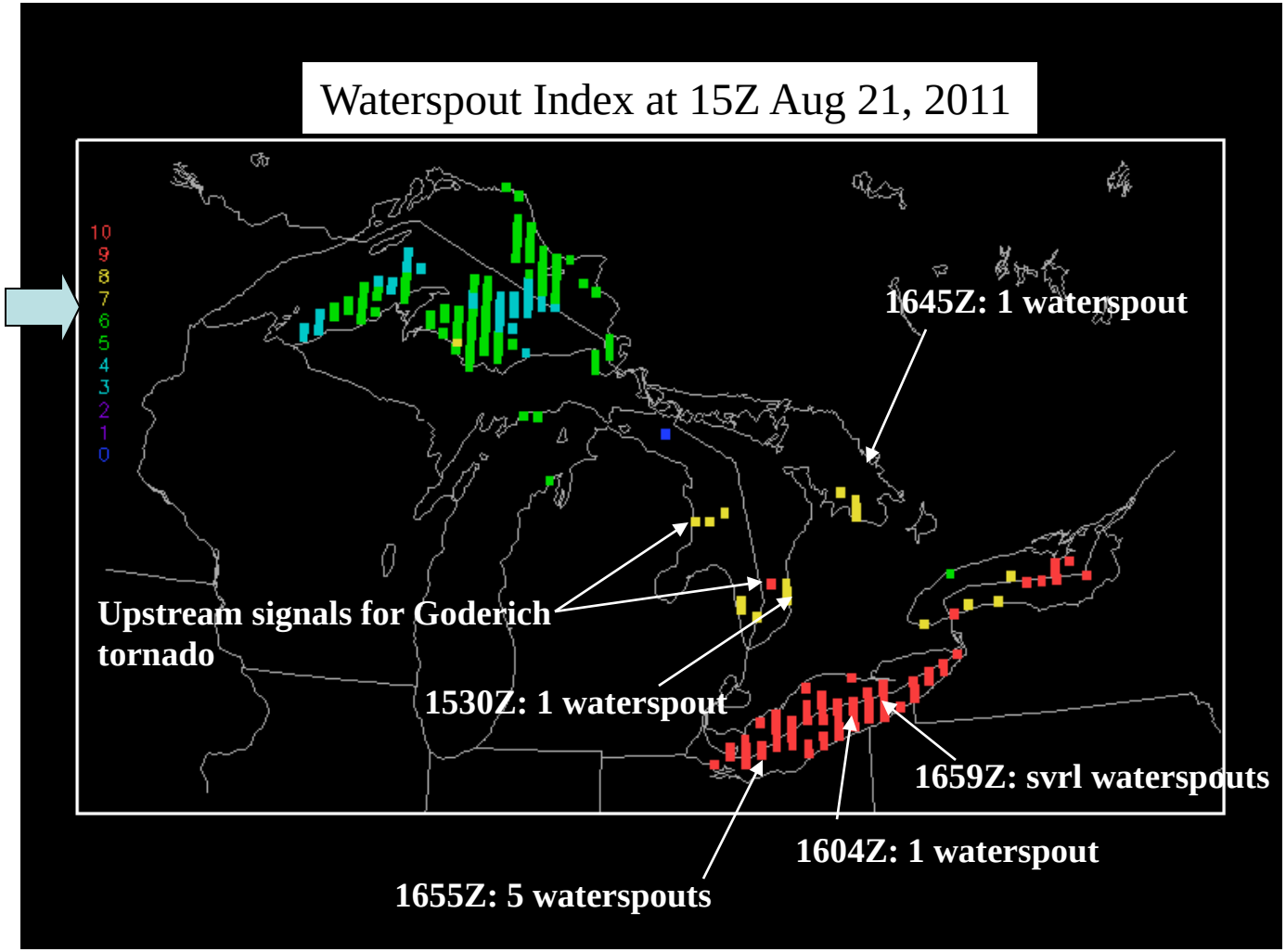
Upstream signals for Goderich waterspout.
Also, several waterspouts reported over Erie



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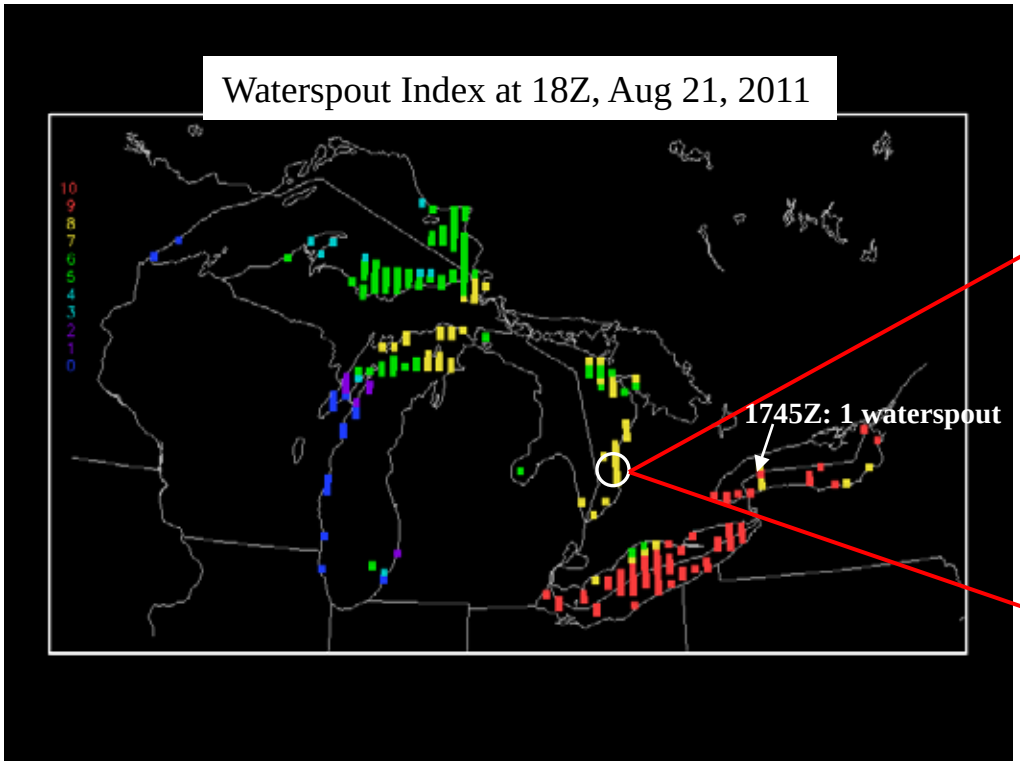
Waterspout spotted at ~1930-1955Z northwest of Goderich



SWI works very well for this event.

http://www.ctv.ca/gallery/html/goderich-tornado/index_.html

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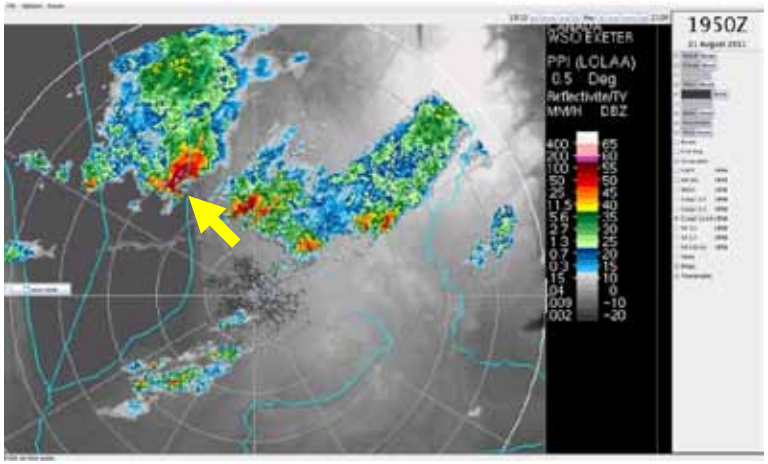
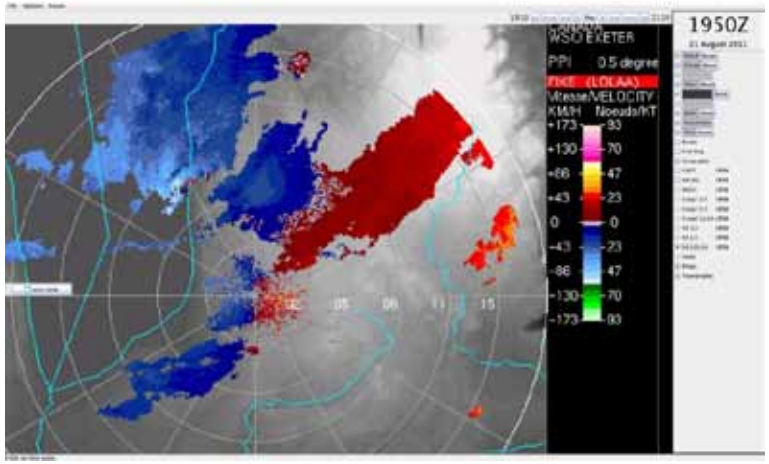
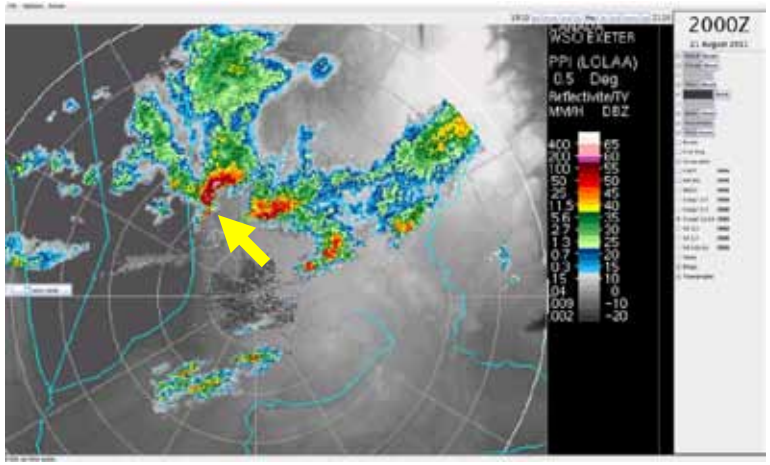
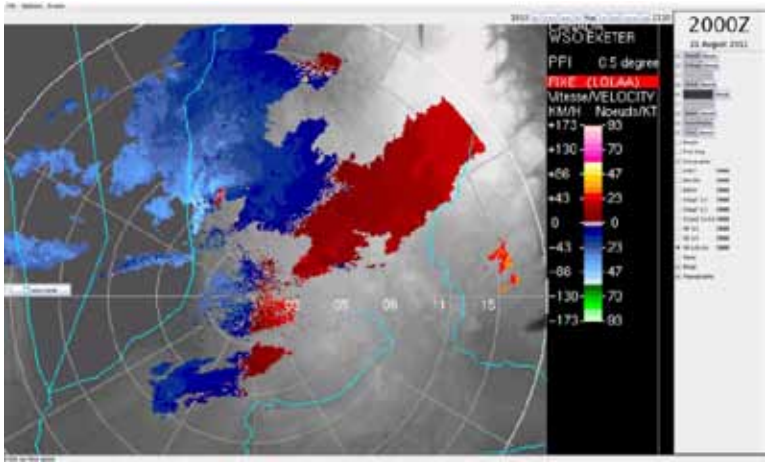
Hook echo were clearly seen from radar at 1950 and 2000Z as the storm cell moves from water to land









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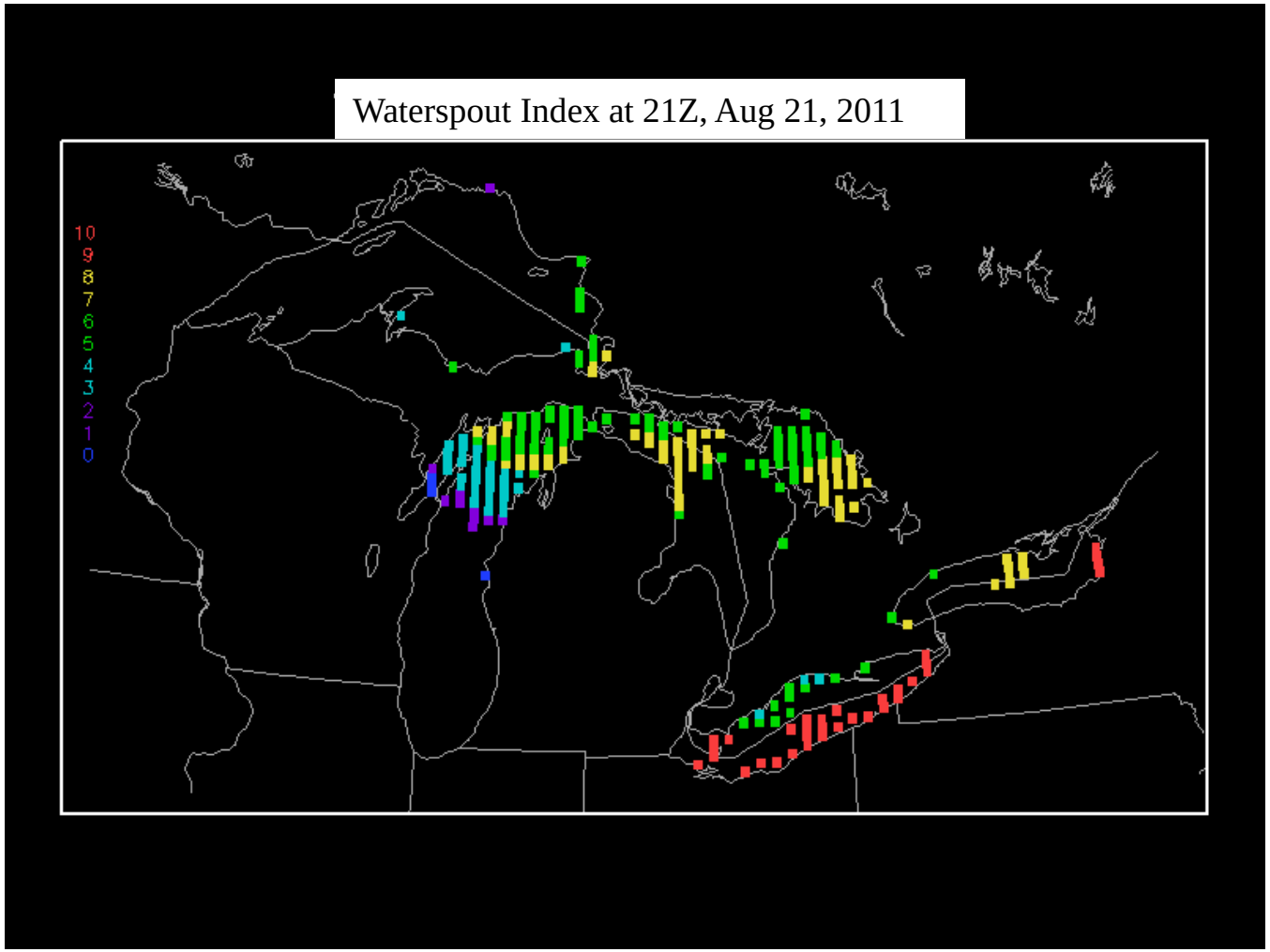
Cold air advection behind front à increasing area of waterspout potential



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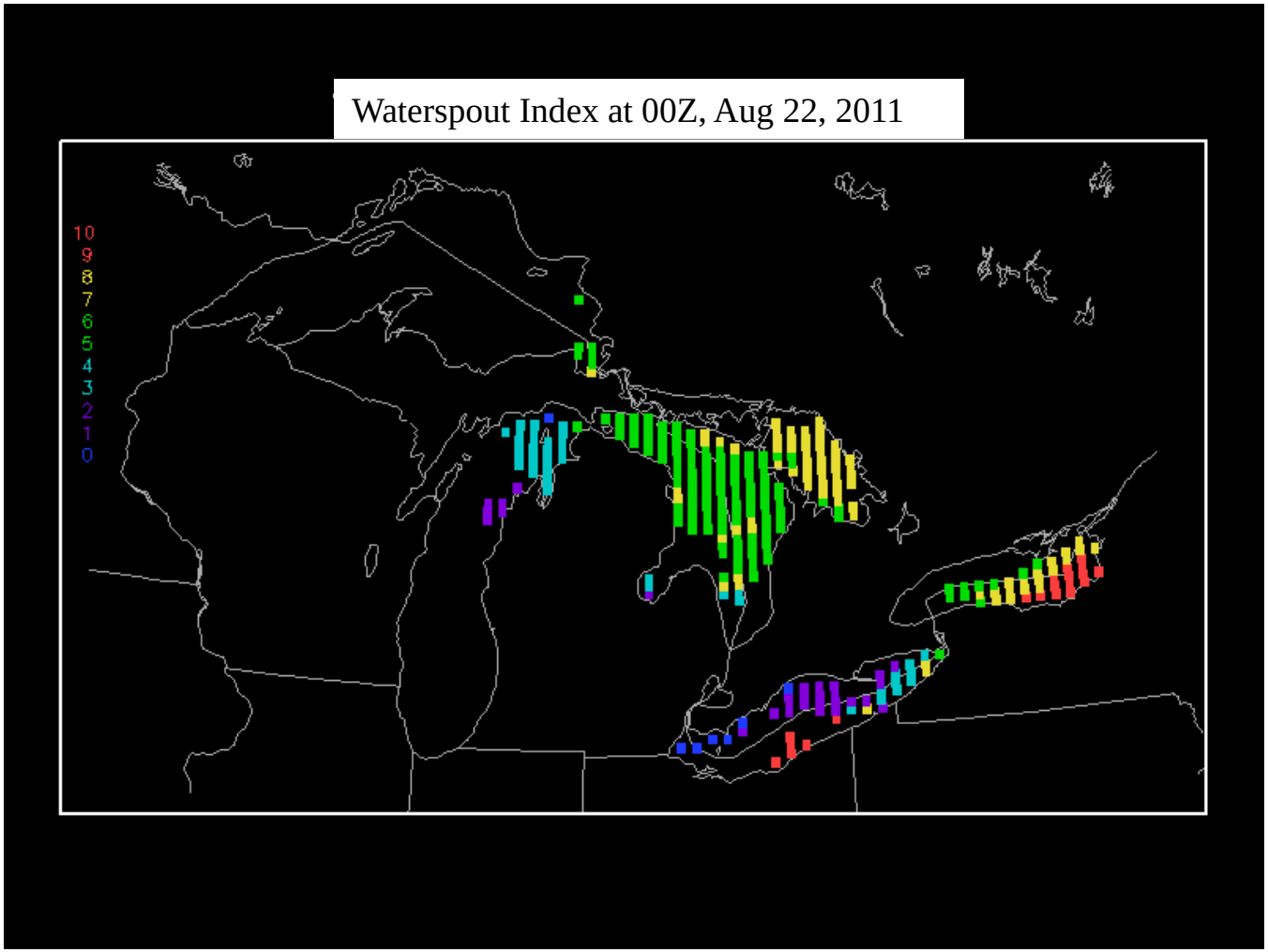
Cold air continues to advect south à area of waterspout potential more extensive



Conclusion

- The new waterspout prognostic system speeds up the process for diagnosing waterspout potential
- The applicability of the algorithm has been demonstrated positively through a number of case studies
- The Goderich case showed that the SWI field could be used as a precursor signal for tornados downstream

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Future Plans

- Adopt a higher resolution grid (0.1 lat x 0.125 long)

- Distinguish between “tornadic” vs “non-tornadic” waterspouts

- Include surface convergent fields (GemReg/GemLam) to Refine the risk area

- Automated output 24/7 this July on OSPC website (http://ospcweb/)

- Expand to other marine areas: Atlantic/Pacific coasts, globally (www.icwr.ca)

- Relate SWI to landspouts

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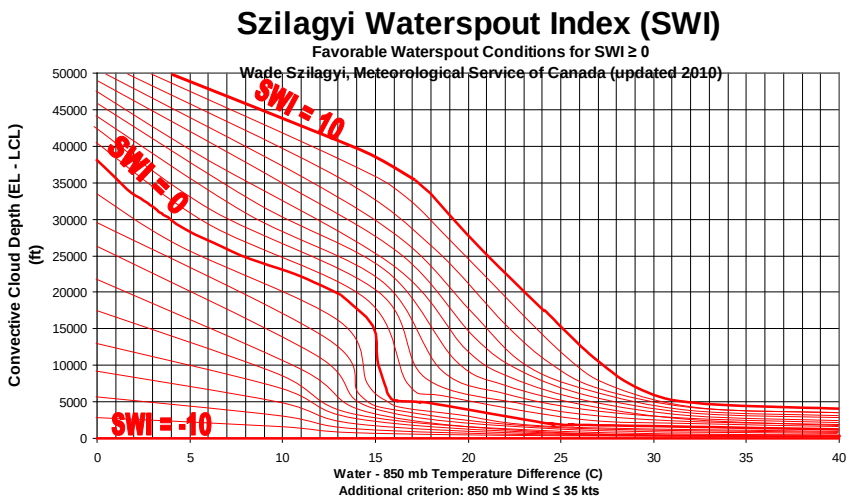
Conversion of the Nomogram to SWI Look Up Table

- Each (ΔT,ΔZ) pair has an associated SWI value

Conversion of nomogram to SWI through a lookup table



SWI Lookup Table

ΔT	Cloud Depth (ΔZ)		SWI
.			
.			
-1	54000	55000	8.5
-1	55000	56000	9
-1	56000	57000	9.5
-1	57000	58000	10
.			
.			
0	3000	4000	-8.5
0	4000	5000	-8
0	5000	6000	-7.5
0	6000	7000	-7
.			
.			
1	9000	10000	-6
1	10000	11000	-5.5
1	11000	12000	-5.5
.			
.			
6	33000	34000	3.5
6	35000	36000	4.5
6	36000	37000	5
6	38000	39000	6
.			
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